GENERAL SCIENCE: BIOLOGY AND PHYSIOLOGY

CELL AND ORGANELLE MAP → MEMBRANE TRANSPORT DECISION TREE → CELL DIVISION COMPARISON → GENE-EXPRESSION RAIL → NUTRITION AND DIGESTION ROUTE → TRANSPORT RESPIRATION EXCRETION LOOP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

★ PRIMARY CORE FLOW · ◇ SUBORDINATE EXTRA · ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00 → STAGE 00

CELL AND ORGANELLE MAP

CELL AND ORGANELLE MAP · NO MEMBRANE-BOUND NUCLEUS OR ORGANELLES · NUCLEUS + COMPARTMENTAL ORGANELLES · ER + GOLGI · PROCESSING AND MOVEMENT · CELLULOSE WALL + CHLOROPLAST + LARGE VACUOLE

PROKARYOTE → no membrane-bound nucleus or organelles

EUKARYOTE → nucleus + compartmental organelles → RIBOSOME → protein → MITOCHONDRION → ATP

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• PROKARYOTE → no membrane-bound nucleus or organelles • EUKARYOTE → nucleus + compartmental organelles	• RIBOSOME → protein • MITOCHONDRION → ATP	• ER + GOLGI → processing and movement • PLANT → cellulose wall + chloroplast + large vacuole

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLANT → cellulose wall + chloroplast + large vacuole.

01 → STAGE 01

MEMBRANE TRANSPORT DECISION TREE

MEMBRANE TRANSPORT DECISION TREE · SOLUTE DOWN GRADIENT · WATER + SELECTIVE MEMBRANE · AGAINST GRADIENT + ENERGY · ACTIVE TRANSPORT · EXTERNAL TONICITY · CELL WATER RESPONSE · MOVEMENT DEPENDS ON MEMBRANE AND GRADIENT

SOLUTE DOWN GRADIENT → diffusion

WATER + SELECTIVE MEMBRANE → osmosis → AGAINST GRADIENT + ENERGY → active transport → EXTERNAL TONICITY → cell water response

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• SOLUTE DOWN GRADIENT → diffusion • WATER + SELECTIVE MEMBRANE → osmosis	• AGAINST GRADIENT + ENERGY → active transport • EXTERNAL TONICITY → cell water response	• TRAP → movement depends on membrane and gradient • SOLUTE DOWN GRADIENT → diffusion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → movement depends on membrane and gradient.

02 → STAGE 02

CELL DIVISION COMPARISON

CELL DIVISION COMPARISON · ONE DIVISION · TWO SIMILAR CELLS · GROWTH AND REPAIR · TWO DIVISIONS · FOUR HAPLOID PRODUCTS · GAMETES + VARIATION · CROSSING OVER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
MITOSIS → one division → two similar cells	MEIOSIS → two divisions → four haploid products	CROSSING OVER != gene editing
ROLE → growth and repair	ROLE → gametes + variation	MITOSIS → one division → two similar cells

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
• MITOSIS → one division → two similar cells • ROLE → growth and repair	• MEIOSIS → two divisions → four haploid products • ROLE → gametes + variation	• CROSSING OVER != gene editing • MITOSIS → one division → two similar cells

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEIOSIS → two divisions → four haploid products.

03 → STAGE 03

GENE-EXPRESSION RAIL

GENE-EXPRESSION RAIL · DNA →REPLICATION→ DNA · DNA →TRANSCRIPTION→ MRNA MRNA + TRNA + RIBOSOME →TRANSLATION→ · POST-TRANSCRIPTIONAL REGULATION · CELL STRUCTURE

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• DNA →replication→ DNA • DNA →transcription→ mRNA mRNA + tRNA + RIBOSOME →translation→ protein microRNA → post-transcriptional regulation	• PROTEIN → cell structure / enzyme / receptor / signal • DNA →replication→ DNA	• DNA →transcription→ mRNA mRNA + tRNA + RIBOSOME →translation→ protein microRNA → post-transcriptional regulation • PROTEIN → cell structure / enzyme / receptor / signal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROTEIN → cell structure / enzyme / receptor / signal.

04 → STAGE 04

NUTRITION AND DIGESTION ROUTE

NUTRITION AND DIGESTION ROUTE · ACID + PEPSIN · FAT EMULSIFICATION · PANCREAS + INTESTINE · DIGESTION COMPLETION · SMALL INTESTINE · PRINCIPAL ABSORPTION

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• MOUTH → amylase • STOMACH → acid + pepsin	• LIVER / BILE → fat emulsification • PANCREAS + INTESTINE → digestion completion	• SMALL INTESTINE → principal absorption • MOUTH → amylase

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PANCREAS + INTESTINE → digestion completion; SMALL INTESTINE → principal absorption.

05 → STAGE 05

TRANSPORT RESPIRATION EXCRETION LOOP

TRANSPORT RESPIRATION EXCRETION LOOP · ALVEOLAR GAS EXCHANGE · GASES + NUTRIENTS + HORMONES + WASTES · PULMONARY + SYSTEMIC CIRCULATION · ATP-PRODUCING RESPIRATION · FILTRATION + REABSORPTION + SECRETION

LUNGS → alveolar gas exchange → BLOOD → gases + nutrients + hormones + wastes → HEART → pulmonary + systemic circulation → CELLS → ATP-producing respiration → NEPHRON → filtration + reabsorption + secretion

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• LUNGS → alveolar gas exchange • BLOOD → gases + nutrients + hormones + wastes	• HEART → pulmonary + systemic circulation • CELLS → ATP-producing respiration	• NEPHRON → filtration + reabsorption + secretion • LUNGS → alveolar gas exchange

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLOOD → gases + nutrients + hormones + wastes.

06 → STAGE 06

CONTROL AND FEEDBACK LADDER

CONTROL AND FEEDBACK LADDER · CHANGE DETECTED · RAPID WIRED RESPONSE · BLOOD-BORNE HORMONE RESPONSE · PHYSIOLOGICAL ADJUSTMENT · NEGATIVE FEEDBACK · RESTORES REGULATED RANGE

RECEPTOR → change detected

NERVOUS → rapid wired response → ENDOCRINE → blood-borne hormone response → TARGET → physiological adjustment

STARTING CONDITION / INPUT	SCIENTIFIC OR ENGINEERING MECHANISM	OUTPUT / FAILURE BOUNDARY
• RECEPTOR → change detected • NERVOUS → rapid wired response	• ENDOCRINE → blood-borne hormone response • TARGET → physiological adjustment	• NEGATIVE FEEDBACK → restores regulated range • RECEPTOR → change detected

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TARGET → physiological adjustment; NEGATIVE FEEDBACK → restores regulated range.

07 → STAGE 07

IMMUNITY ARCHITECTURE

IMMUNITY ARCHITECTURE · BARRIERS + PHAGOCYTES + COMPLEMENT · B CELL · T CELL · REGULATORY ROLES · REGULATORY T CELL + FOXP3 · PERIPHERAL TOLERANCE · SELF-REACTIVITY CONTROL

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• BARRIERS + PHAGOCYTES + COMPLEMENT → innate • B CELL → antibody / memory	• T CELL → helper / cytotoxic / regulatory roles • REGULATORY T CELL + Foxp3 → peripheral tolerance	• SELF-REACTIVITY CONTROL != no immune response • BARRIERS + PHAGOCYTES + COMPLEMENT → innate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REGULATORY T CELL + Foxp3 → peripheral tolerance.

08 → STAGE 08

DISEASE TRANSMISSION GRID

DISEASE TRANSMISSION GRID · CARRIES PATHOGEN, IS NOT DISEASE · ANIMAL-HUMAN INTERFACE · MECHANISM-SPECIFIC LAYERS

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
VECTOR → carries pathogen, is not disease	PREVENTION → mechanism-specific layers	ZOONOSIS → animal-human interface
ZOONOSIS → animal-human interface	VECTOR → carries pathogen, is not disease	PREVENTION → mechanism-specific layers

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• VECTOR → carries pathogen, is not disease • ZOONOSIS → animal-human interface	• PREVENTION → mechanism-specific layers • VECTOR → carries pathogen, is not disease	• ZOONOSIS → animal-human interface • PREVENTION → mechanism-specific layers

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AIR | WATER/FOOD | BLOOD | CONTACT | VECTOR.

09 → STAGE 09

MICROBIAL ECOLOGY AND RESISTANCE

MICROBIAL ECOLOGY AND RESISTANCE · ANTIMICROBIAL PRESSURE · RESISTANT MICROBE · SURVIVAL + SPREAD · MATRIX-PROTECTED COMMUNITY · CONTEXT-DEPENDENT ROLES · EXTREMOPHILE LABEL · TOLERANCE, NOT UNIVERSAL SAFETY

ANTIMICROBIAL PRESSURE → selection

RESISTANT MICROBE → survival + spread → BIOFILM → matrix-protected community → MICROBIOME / PROBIOTIC → context-dependent roles

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• ANTIMICROBIAL PRESSURE → selection • RESISTANT MICROBE → survival + spread	• BIOFILM → matrix-protected community • MICROBIOME / PROBIOTIC → context-dependent roles	• EXTREMOPHILE LABEL → tolerance, not universal safety • MUST REMEMBER: Biology and physiology connect cell structure, metabolism, genetics,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Biology and physiology connect cell structure, metabolism, genetics,.

10 → STAGE 10

PLANT FUNCTION FLOW

PLANT FUNCTION FLOW · WATER + MINERAL UPTAKE · UPWARD TRANSPORT · STOMATA + PHOTOSYNTHESIS + TRANSPIRATION · FOOD TRANSLOCATION · DIFFERENT CARBON-FIXATION STRATEGIES · CLOSE DISTINCTION · GENE IS NOT ALLELE OR GENOME, GENOTYPE IS NOT

ROOT → water + mineral uptake → XYLEM → upward transport → LEAF → stomata + photosynthesis + transpiration → PHLOEM → food translocation → CAM → different carbon-fixation strategies

STARTING CONDITION / INPUT	SCIENTIFIC OR ENGINEERING MECHANISM	OUTPUT / FAILURE BOUNDARY
• ROOT → water + mineral uptake • XYLEM → upward transport	• LEAF → stomata + photosynthesis + transpiration • PHLOEM → food translocation	• CAM → different carbon-fixation strategies • CLOSE DISTINCTION: Gene is not allele or genome, genotype is not phenotype, pathogen is...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Gene is not allele or genome, genotype is not phenotype, pathogen is.

11 → STAGE 11

PYQ AND EVIDENCE-STATUS RAIL

PYQ AND EVIDENCE-STATUS RAIL · PLANT FORM · SPECIES PROPERTY · EXACT CLASSIFICATION · MICROBIAL METABOLISM · PROGRAMME DELIVERY · WHO VALIDATION · PATHOGEN ERADICATION

TAXON / PLANT FORM / SPECIES PROPERTY → exact classification → CELLULOSE → cellulase → sugars → microbial metabolism → RESEARCH → SURVEILLANCE → PROGRAMME DELIVERY → REGULATION → WHO VALIDATION != pathogen eradication → MECHANISM / ASSOCIATION / APPROVAL / OUTCOME → separate rungs

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• TAXON / PLANT FORM / SPECIES PROPERTY → exact classification • CELLULOSE → cellulase → sugars → microbial metabolism	• RESEARCH → SURVEILLANCE → PROGRAMME DELIVERY → REGULATION • WHO VALIDATION != pathogen eradication	• MECHANISM / ASSOCIATION / APPROVAL / OUTCOME → separate rungs • EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name organism/cell/tissue/organ,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name organism/cell/tissue/organ,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name organism/cell/tissue/organ,.

E → EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · BIOTECHNOLOGY POLICY AND PROGRAMME INSTITUTION LINKING BASIC BIOLOGY TO · MOLECULAR BIOLOGY, GENOMICS AND SYSTEMS-BIOLOGY RESEARCH ECOSYSTEMS · DISEASE-SURVEILLANCE AND PUBLIC-HEALTH INSTITUTIONS · WHERE PHYSIOLOGY, MICROBIOLOGY AND IMMUNOLOGY BECOME POLICY ACTION · SEPARATE THE FUNCTIONS PRECISELY · ICMR PERFORMS AND FUNDS BIOMEDICAL RESEARCH · NCDC AND THE IDSP/IHIP NETWORK UNDER MOHFW RUN OPERATIONAL

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- DBT: biotechnology policy and programme institution linking basic biology to applied missions.
- CSIR-CCMB / CSIR-IGIB: molecular biology, genomics and systems-biology research ecosystems.
- ICMR / disease-surveillance and public-health institutions: where physiology, microbiology and immunology become policy action.
- CAUTION: Separate the functions precisely: ICMR performs and funds biomedical research

READINESS, UNIT OR STATUS LIMIT

- NCDC and the IDSP/IHIP network under MoHFW run operational surveillance and outbreak response
- State health departments deliver programmes
- CDSCO grants regulatory approvals.
- Research capacity, surveillance capacity and regulatory capacity fail in different ways and require different remedies.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- WHO-linked surveillance and immunization reporting frameworks: relevant because public-health biology is continuously monitored, not studied once and forgotten.
- CAUTION: Applied extensions sit in 13 Biotechnology-Fundamentals-and-DBT-Missions.md, 14 Genetic-Engineering-GM-Crops-and-CRISPR.md and 15 Vaccines-Monoclonal-Antibodies-and-Biopharma.md.
- Genetics and cell biology underpin newborn screening, cancer diagnostics, pathogen surveillance and precision medicine.
- Physiology underlies nutrition programmes, anemia policy, non-communicable disease prevention and occupational health.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels